

Why TMR SENSORS Lead Next-Generation Design

TMR sensors are gaining traction in industries needing precision and power efficiency. What makes them the go-to choice for modern designs?

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Have you noticed the buzz around tunnelling magnetoresistance (TMR) sensors? These advanced sensors are capturing attention among designers seeking unmatched performance, reliability, and precision for demanding applications.

Here is what sets them apart: TMR sensors deliver lower power consumption, superior thermal stability, enhanced resolution, and exceptional sensitivity—key advantages for industries such as industrial automation, electric vehicles (EVs), and high-power systems.

Concerned about durability under tough conditions? Rest assured, TMR sensors are engineered for high-bandwidth applications, enabling them to handle rapid fluctuations and challenging environments with ease. Newer models also feature compact designs, thanks to back-end-of-line (BEOL) manufacturing compatibility.

But how do they compare to traditional

Hall-effect sensors? The difference is significant. TMR sensors leverage changes in electrical resistance to achieve greater sensitivity and stability, making them a standout choice for applications where precision and efficiency are non-negotiable.

From automotive and renewable energy to heavy-duty industrial automation, TMR sensors are fast becoming the preferred solution for industries requiring accurate, power-efficient, and durable sensing technologies.

Looking to enhance accuracy, save energy, and tackle environmental challenges—all within a compact design? TMR sensors are paving the way for today's designs and tomorrow's innovations.

TMR sensors in EVs

Imagine being at the forefront of EV innovation, where cleaner, lighter, and more efficient designs are the goal. In battery monitoring and wireless charging, new closed-loop TMR sensors are driving practical advancements—and they might be exactly what you are seeking.

Why consider TMR? Traditional current sensors rely on a large magnetic flux coil, adding weight and limiting design options. The TMR sensor, with its coreless design, overcomes these challenges. Without needing a coil, it still achieves precise battery monitoring, enabling lighter EVs and greater design flexibility.

The TMR sensor integrates the magnetic sensing component and an application-specific integrated circuit (ASIC) in a single package. What does that mean for you? It means this sensor can measure high currents—up to 1200A—with an error

